

ECSE-4040

Digital Electronics

Lecture 27

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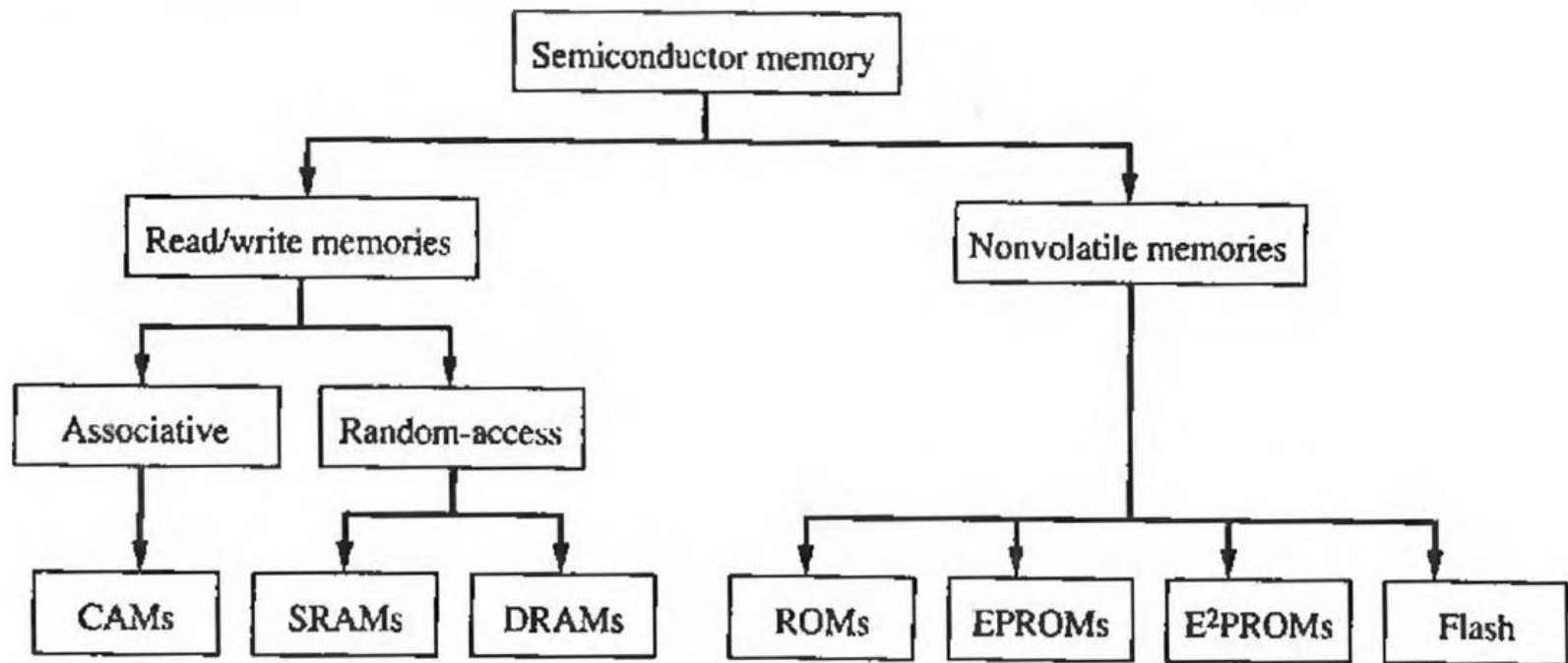
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<https://lms.rpi.edu>

Lecture Outline

- Semiconductor Memories
 - Types
 - Decoders
 - Sense Amplifiers
 - EEPROM and Flash Memories

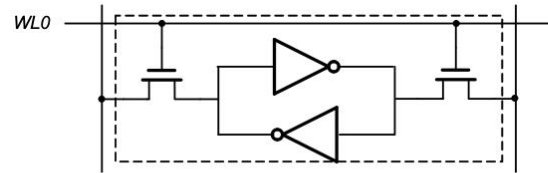
Semiconductor Memories



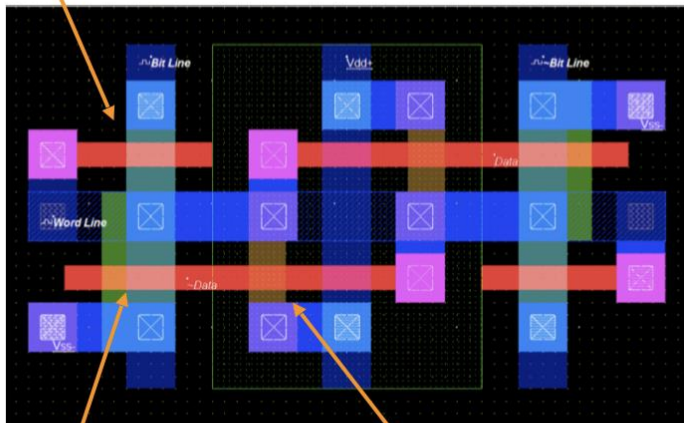
SRAM Cell

Modern SRAM

□ ST/Philips/Motorola

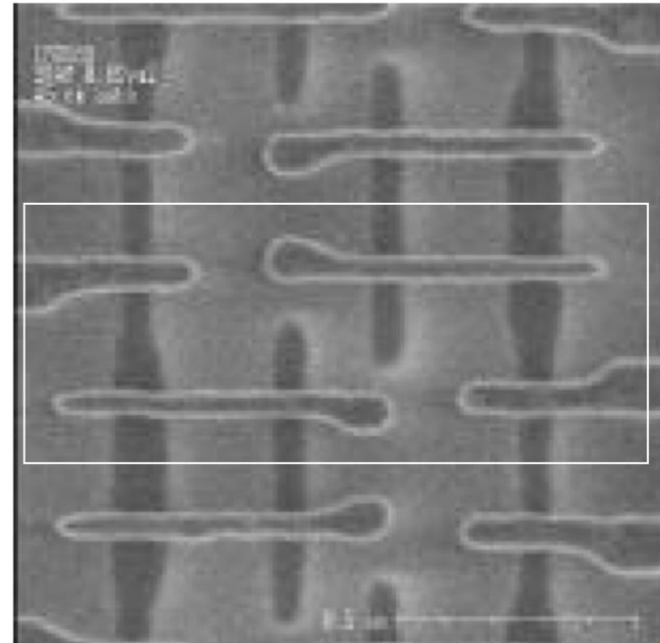


Access Transistor

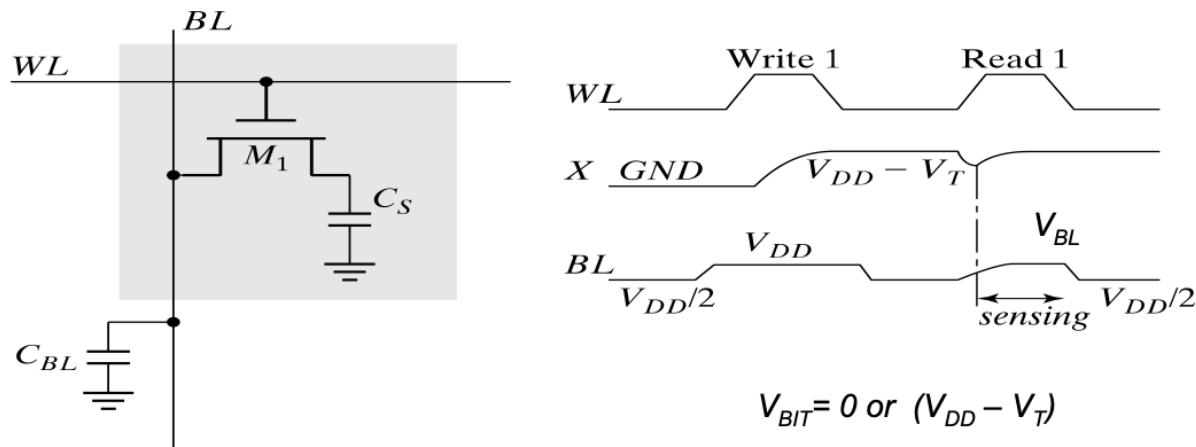


Pull down

Pull up



DRAM Operation



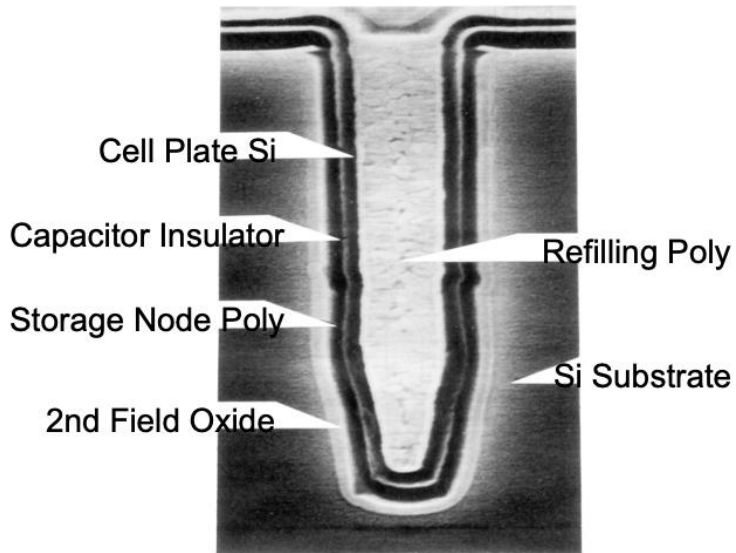
Write: C_S is charged or discharged by asserting WL and BL.

Read: Charge redistribution takes places between bit line and storage capacitance

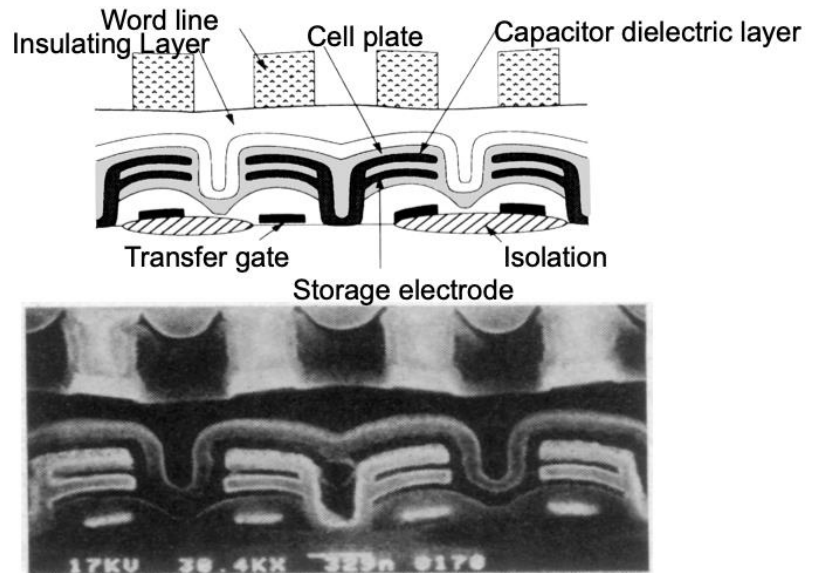
$C_S \ll C_{BL}$ Voltage swing is small; typically around 250 mV.

- ❑ To get sufficient C_S , **special IC process is used**
- ❑ Cell reading is destructive, therefore read operation always is followed by a write-back
- ❑ Cell loses charge (leaks away in ms - highly temperature dependent) therefore cells occasionally need to be “refreshed” - read/write cycle

DRAM Cell

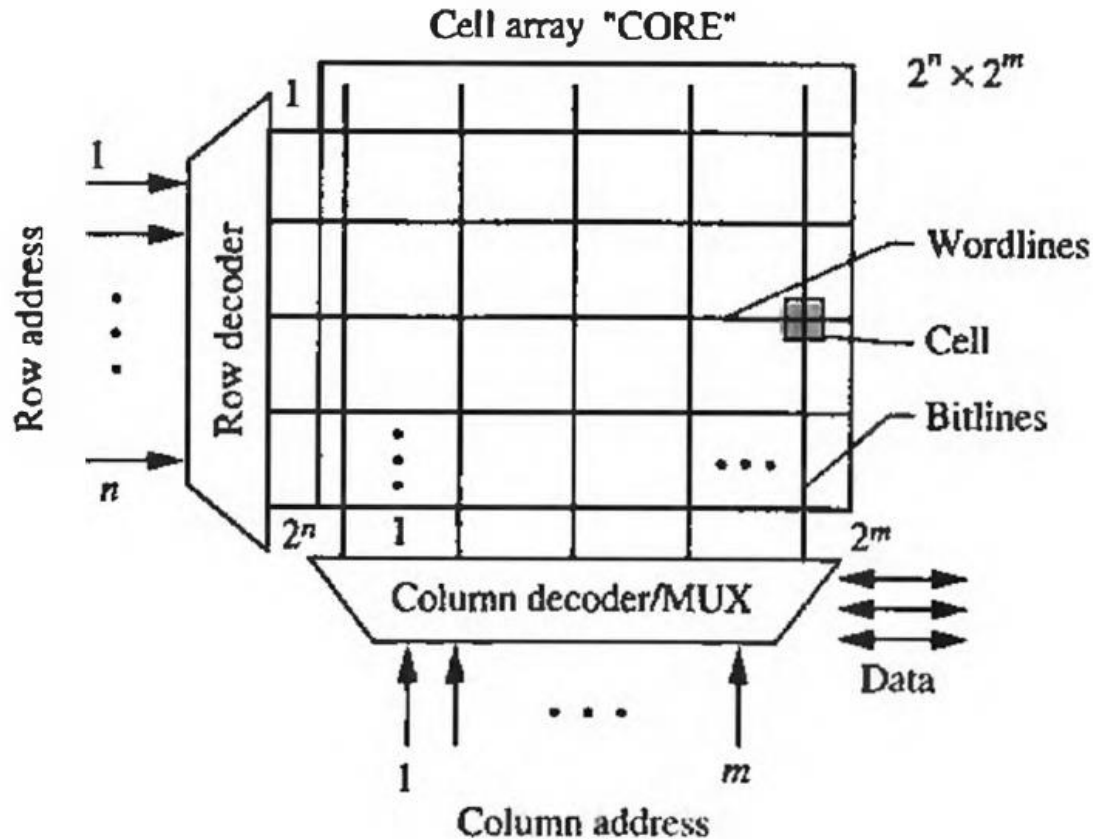


Trench Cell

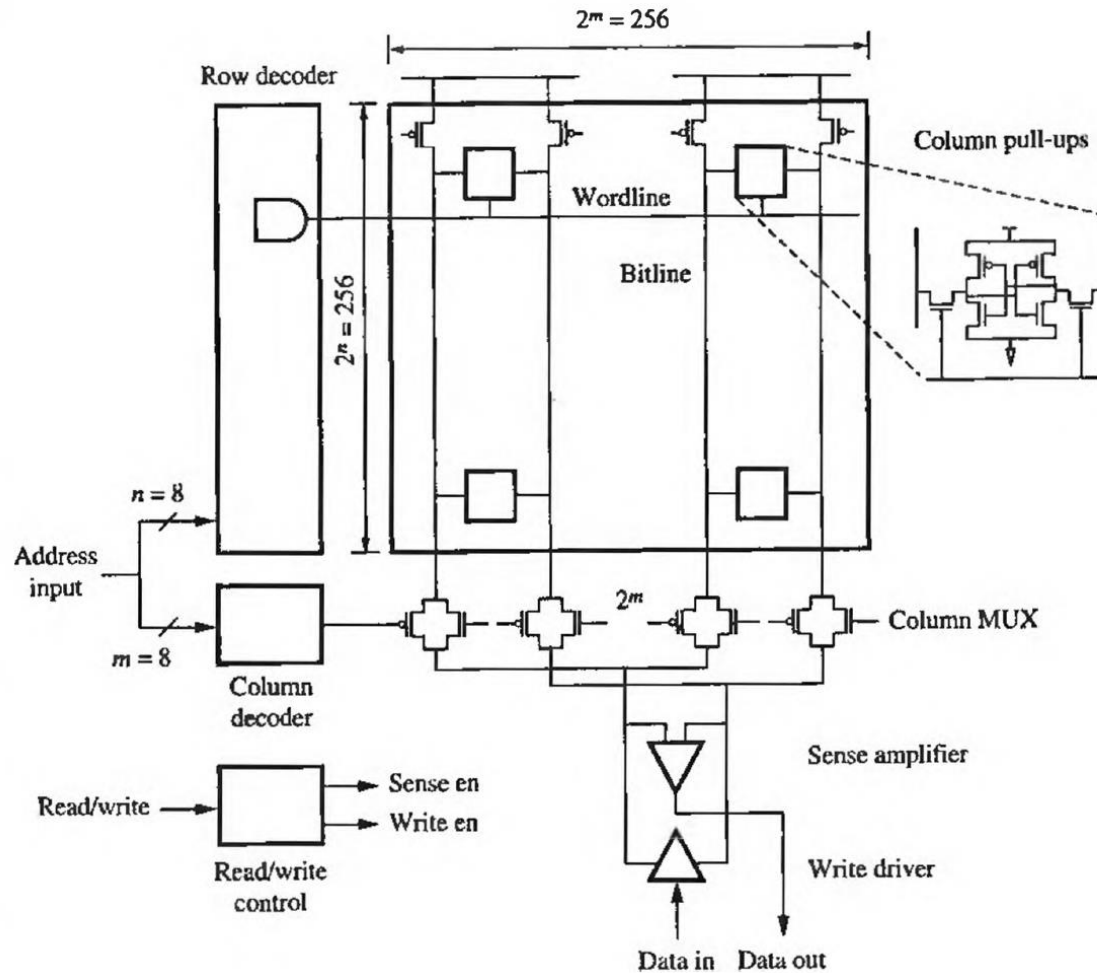


Stacked-capacitor Cell

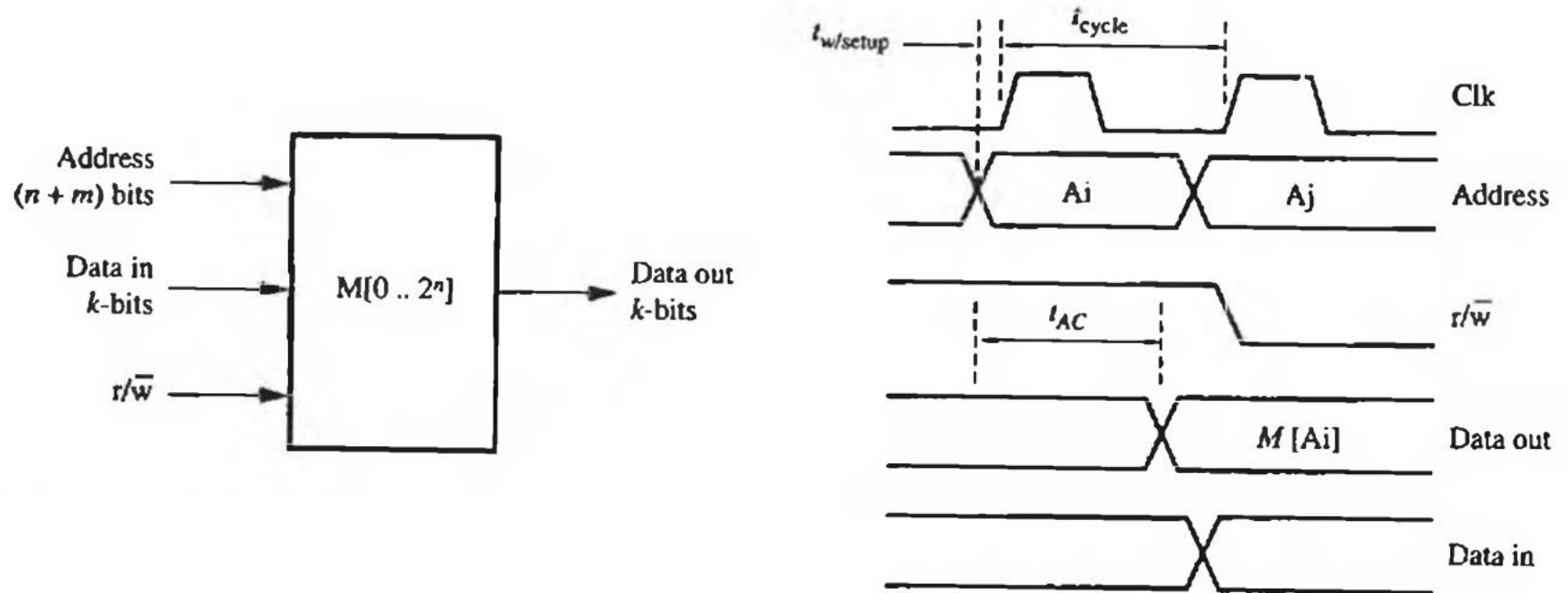
Memory Organization



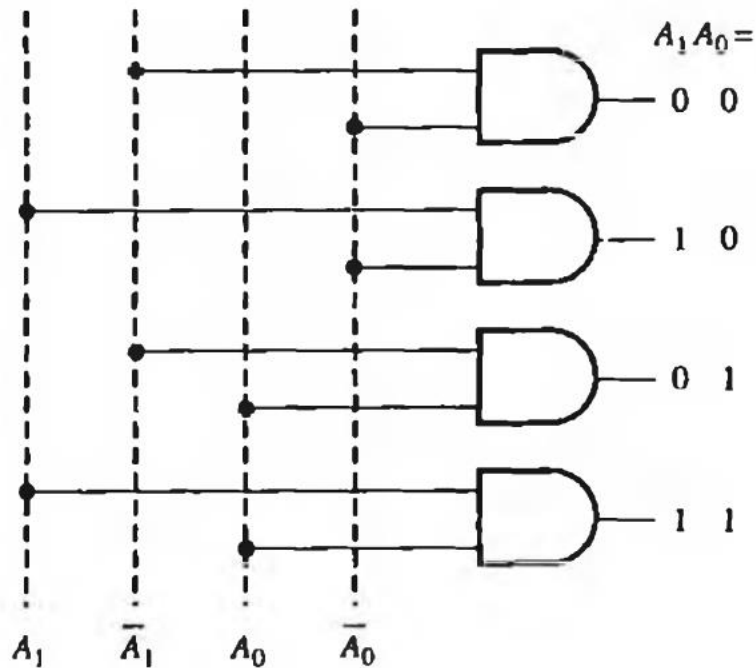
Memory Architecture



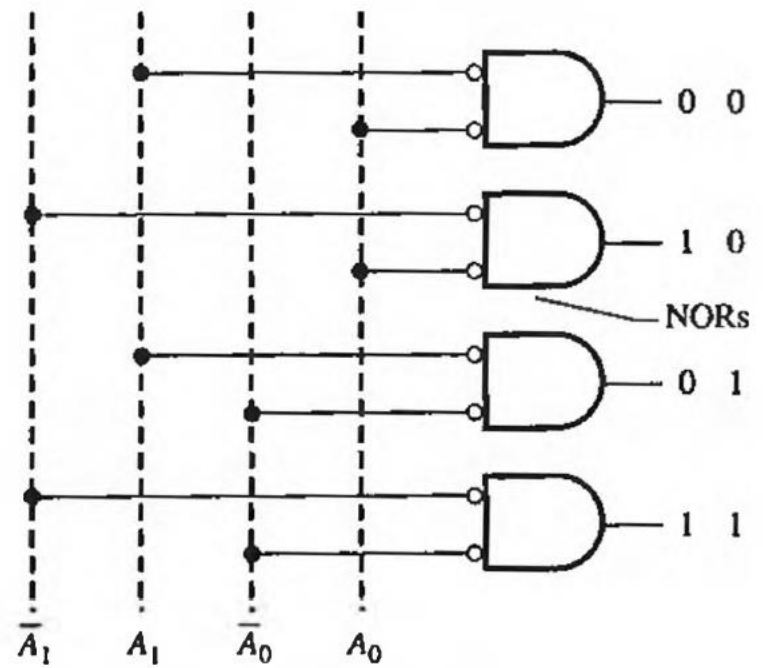
RAM Timing Parameters



MOS Decoders

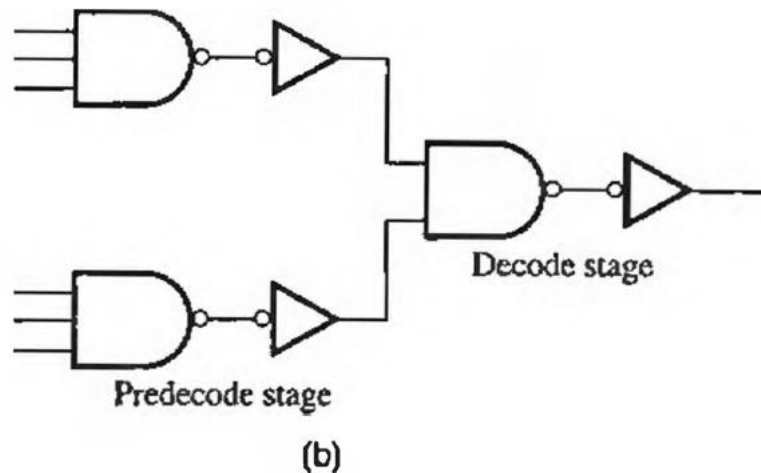
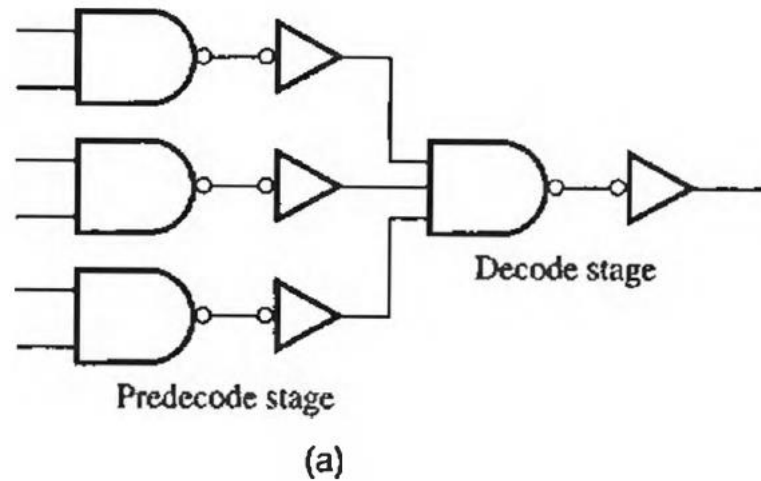


(a) AND-based decoder

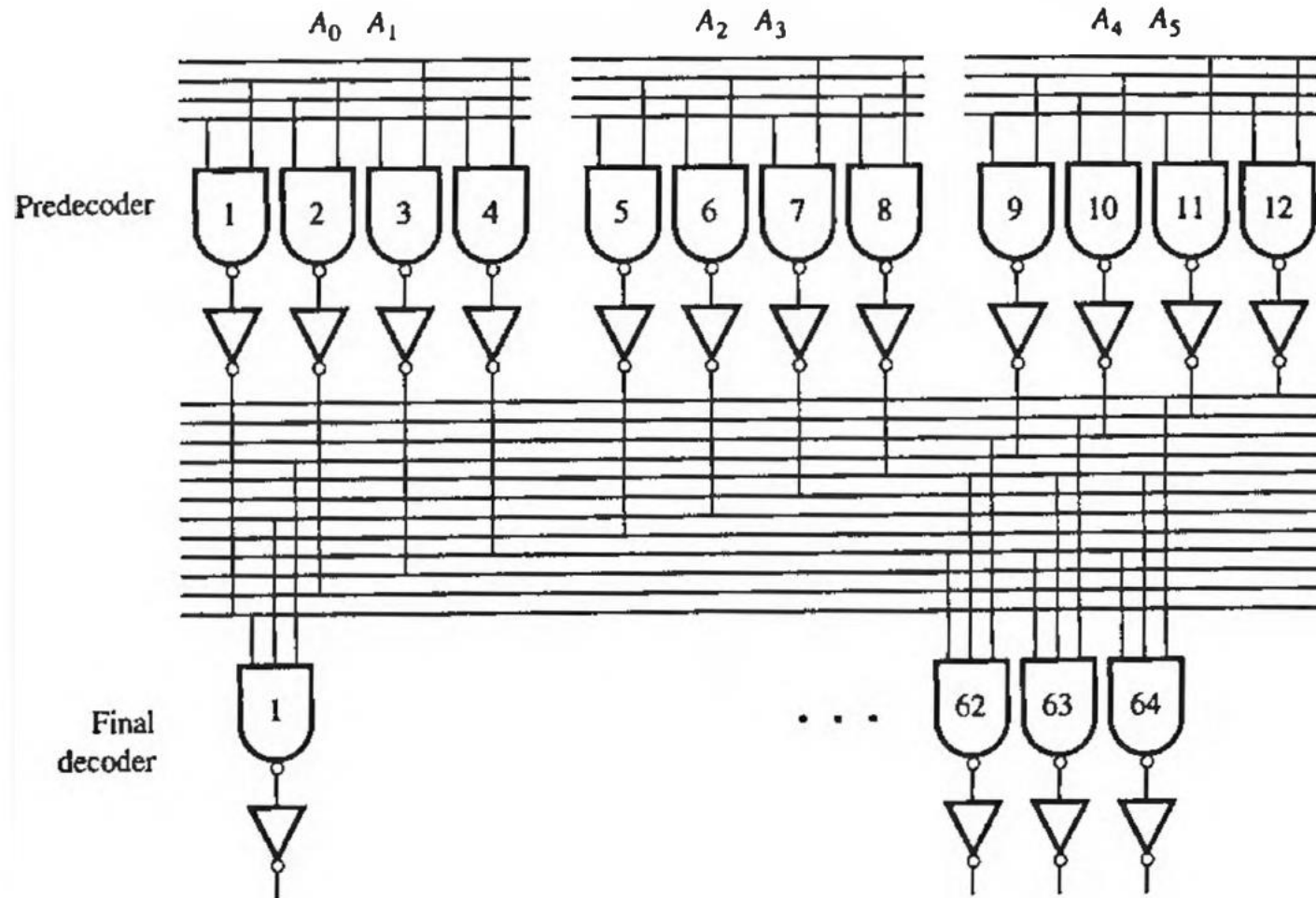


(b) NOR-based decoder

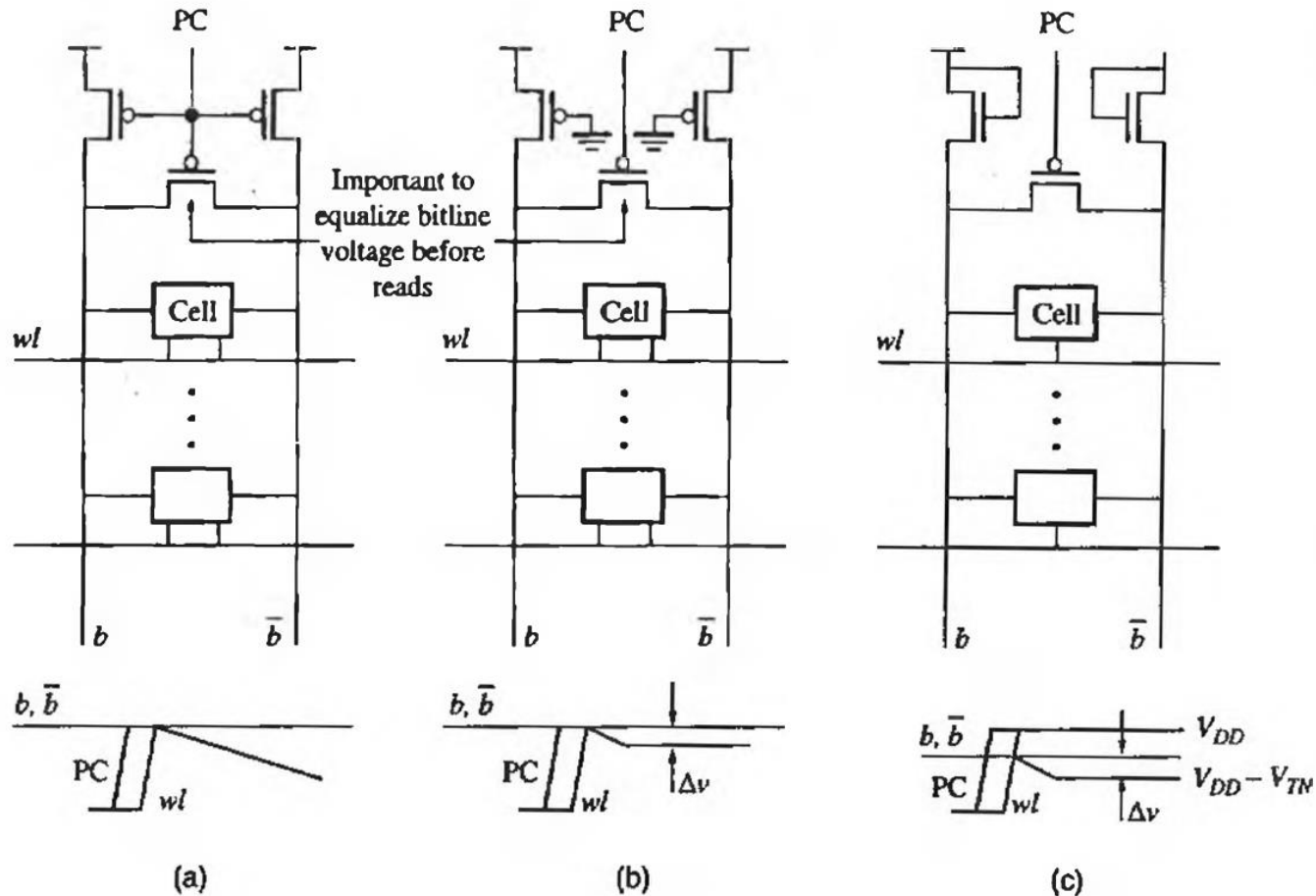
Predecoder Configurations



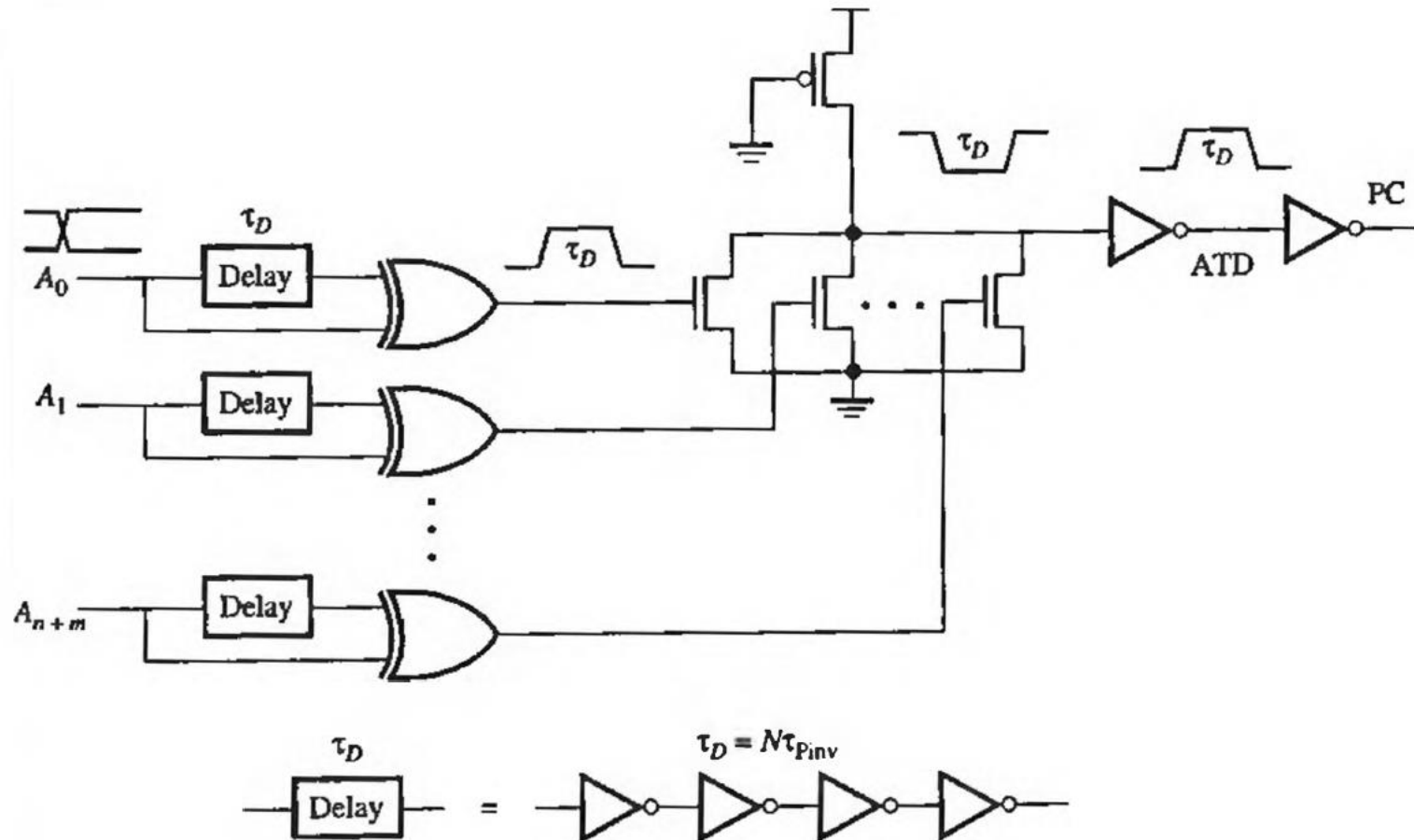
2-Level Decoder



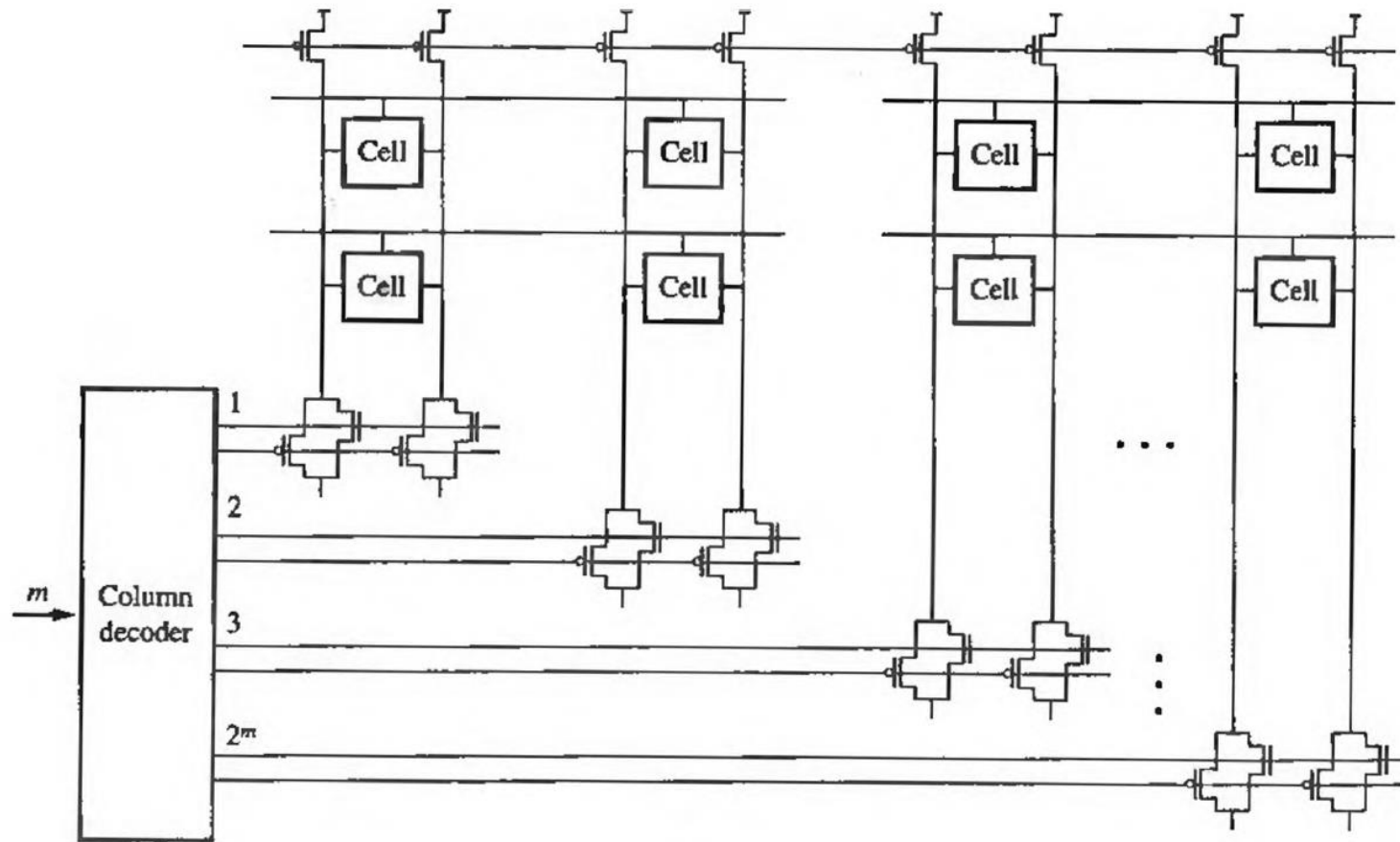
Column Pull-Up Configurations



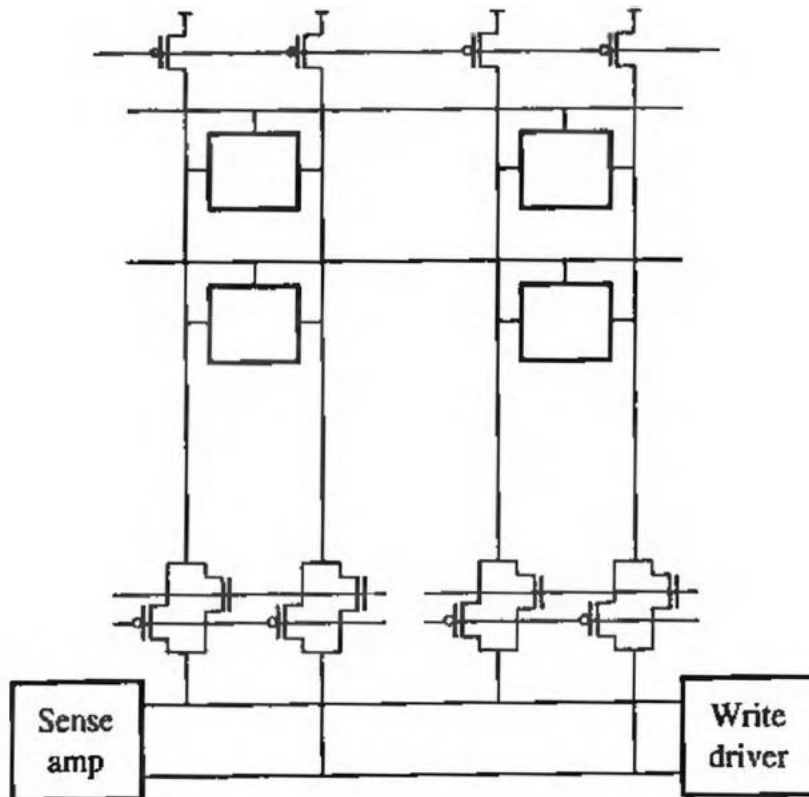
Address Transition Detection Ckt.



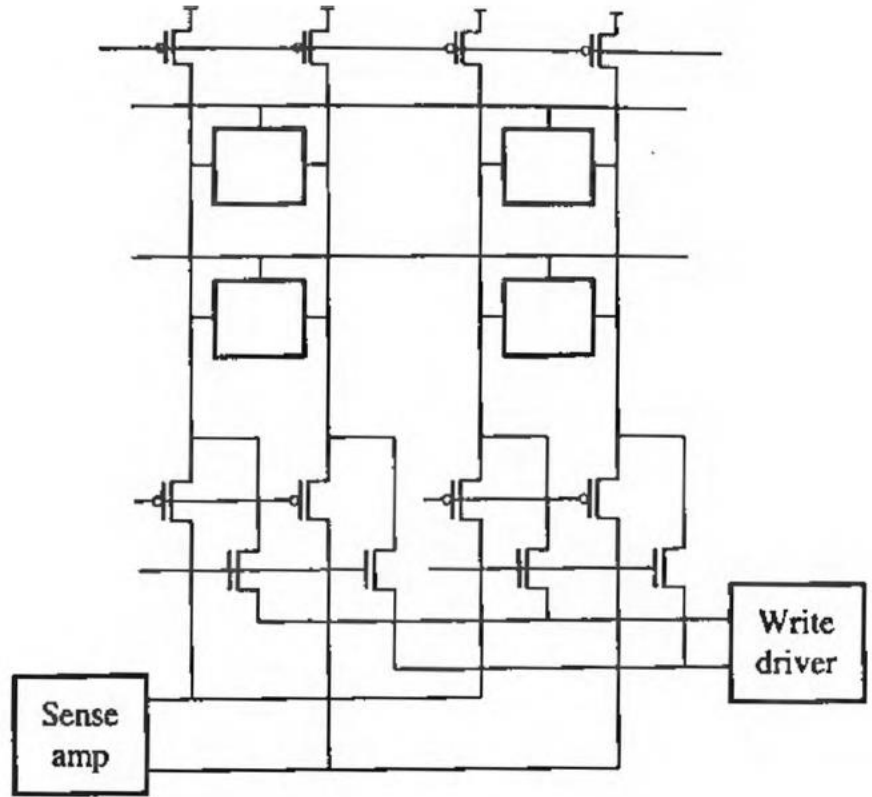
Column Decoding and Multiplexing



Column Selection



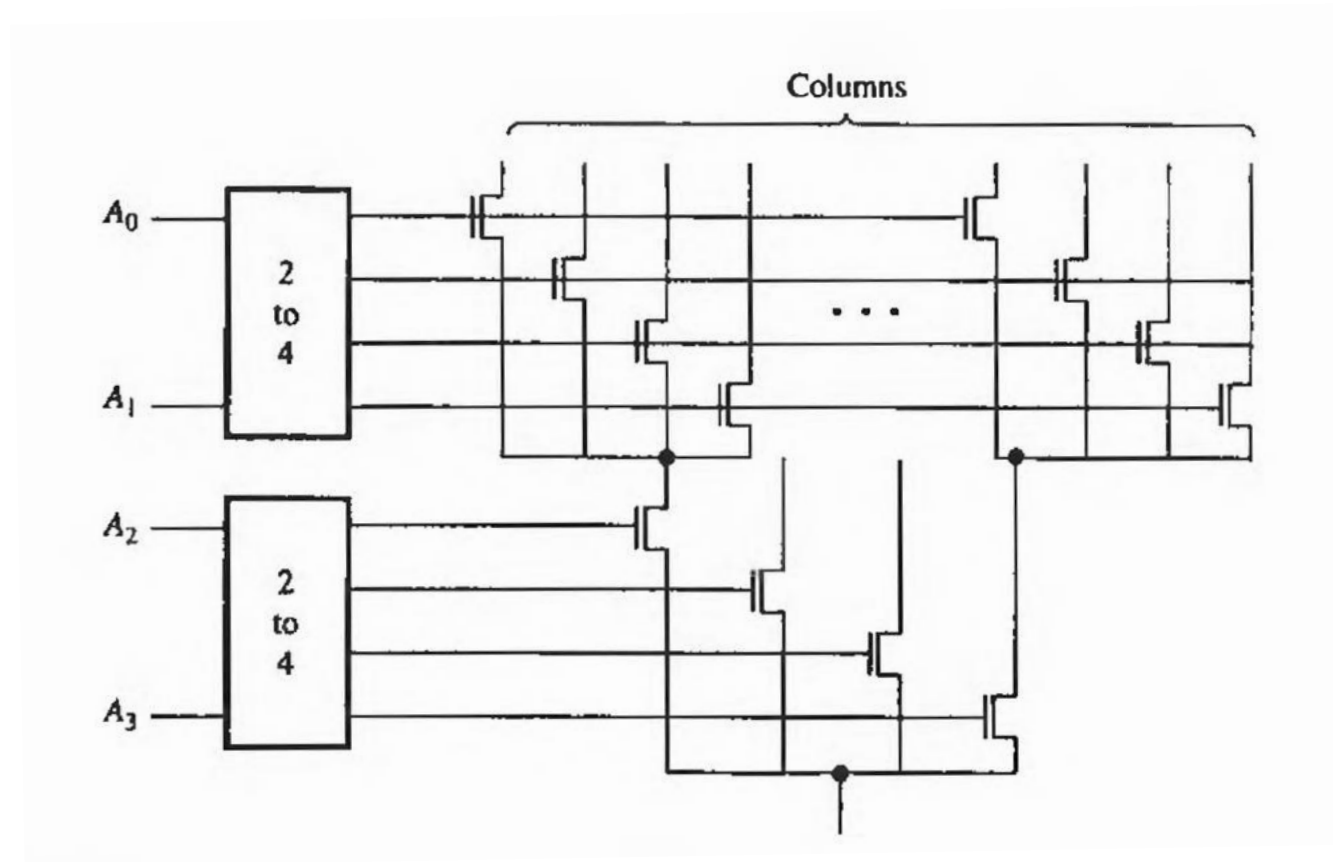
(a)



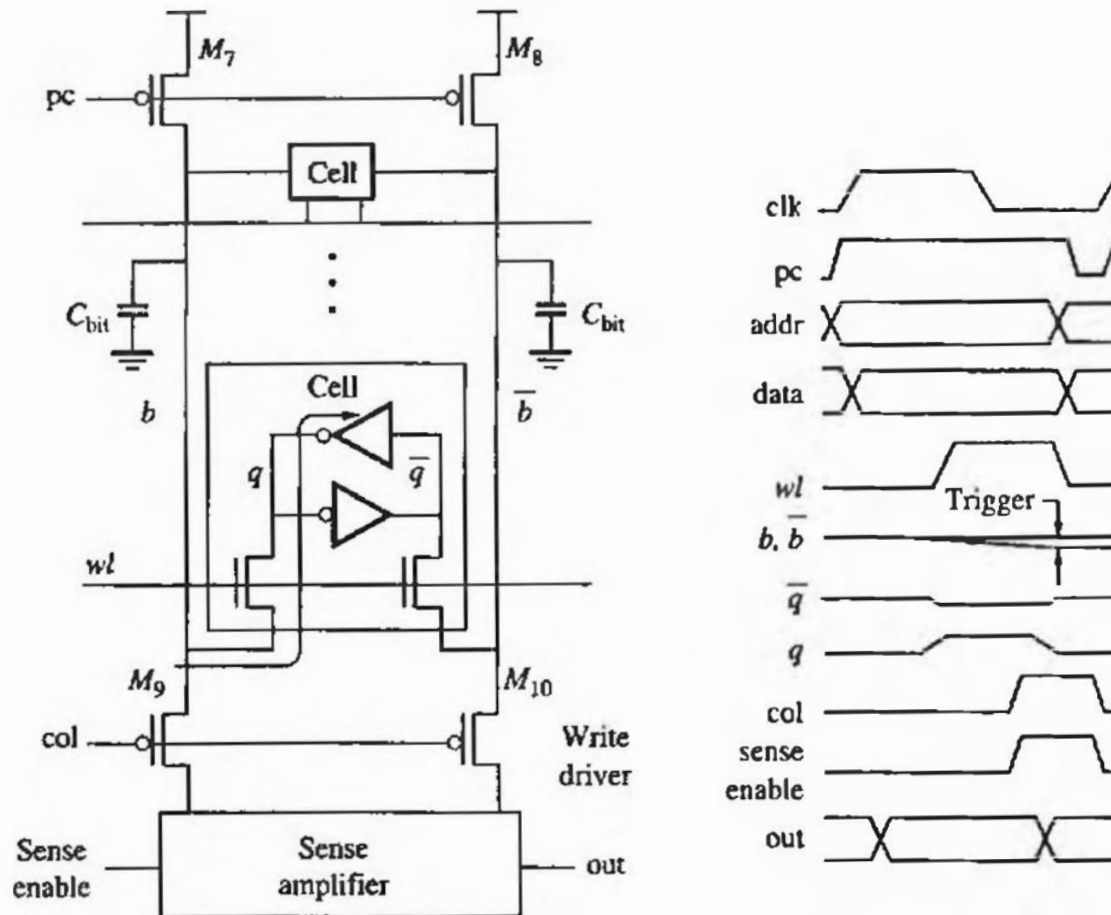
(b)

Decoders

Two-level Tree Decoder



SRAM Read Circuitry



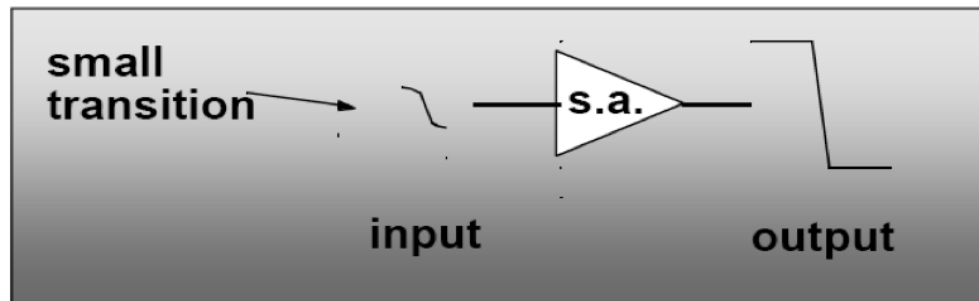
Sense Amplifier

$$\tau_p = \frac{C \cdot \Delta V}{I_{av}}$$

Annotations for the equation:

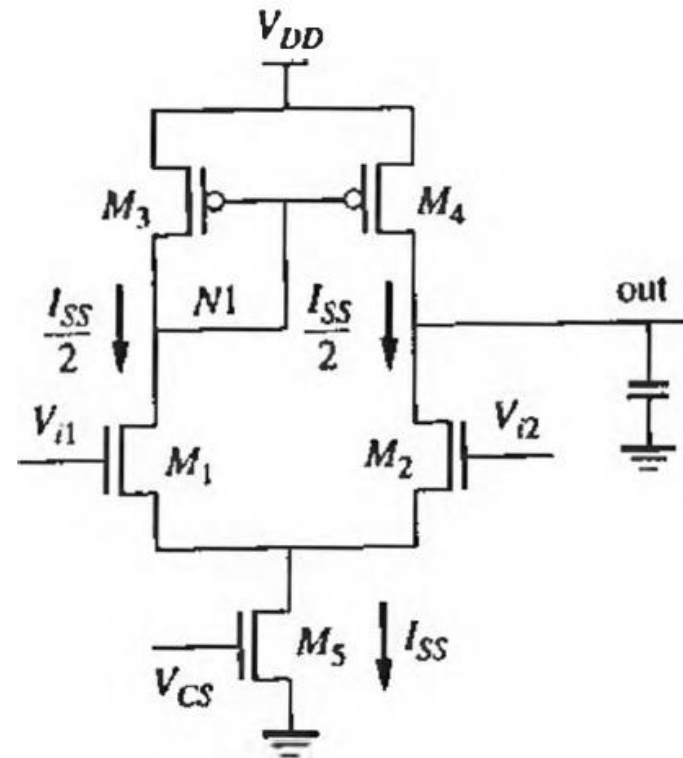
- large → C
- make as small as possible → ΔV
- small → I_{av}

Idea: Use Sense Amplifier



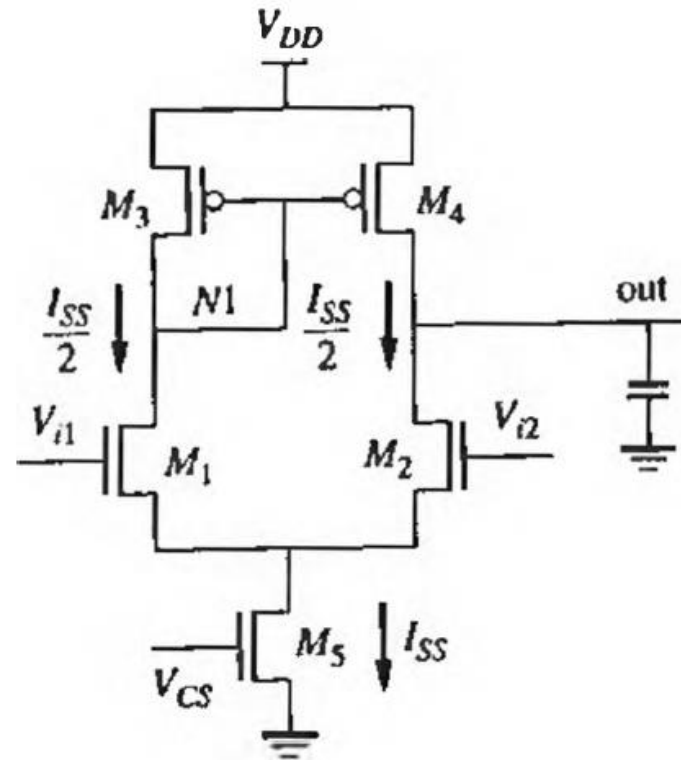
Sense Amplifiers

- Attenuates *common-mode* noises and amplifies *differential-mode* signals
- Improve noise immunity and speed
- Needs to work with saturated enhancement load pull-up transistors

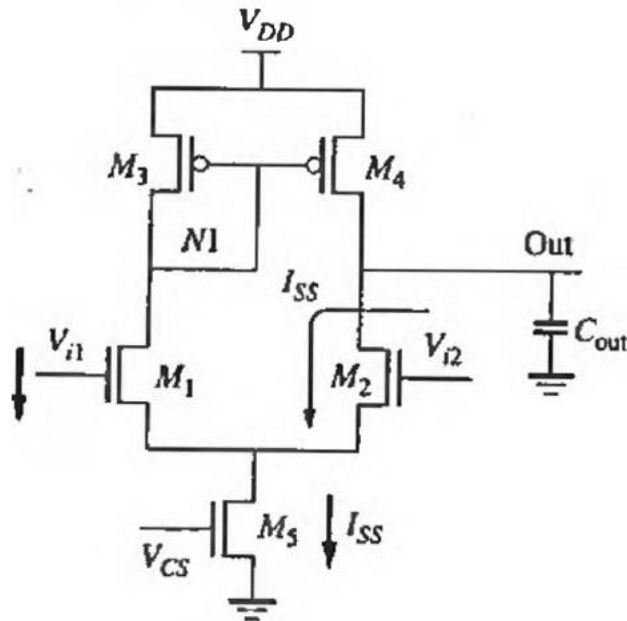


Sense Amplifiers

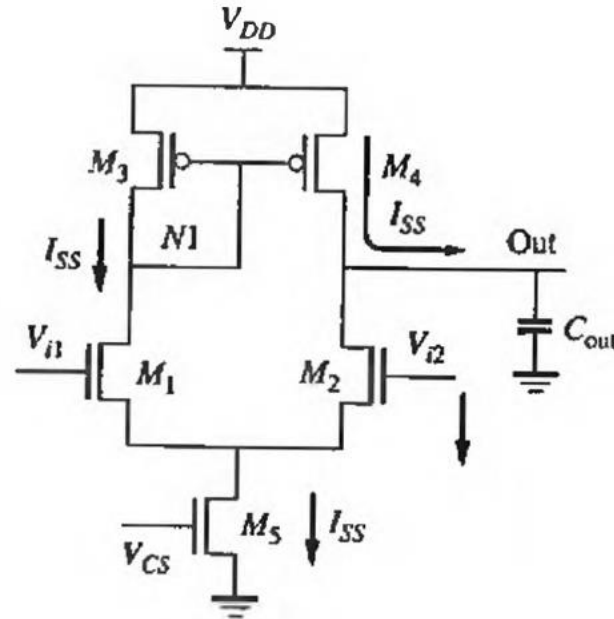
- $I_3 \approx I_4$
- M_1, M_2 must be saturated
so V_{i1} and V_{i2} is set to
 $V_{DD} - V_{TN}$ instead of V_{DD}
- $V_{N1} = V_{out} \approx (V_{DD} - V_{TN}) - V_{TN}$
 $\approx V_{DD} - 2V_{TN}$
to prevent M_3, M_4 to be
cut off



Sense Amplifiers



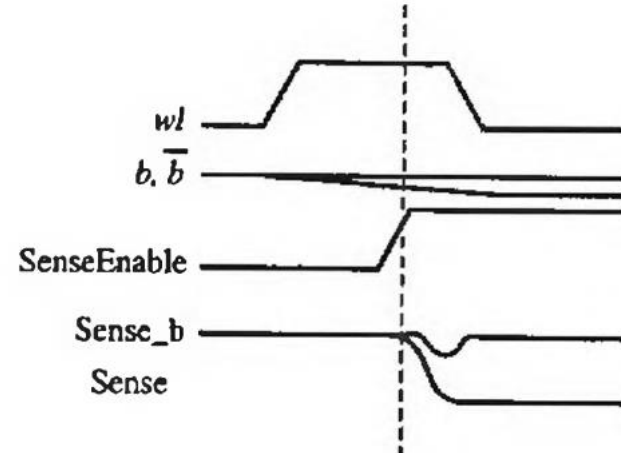
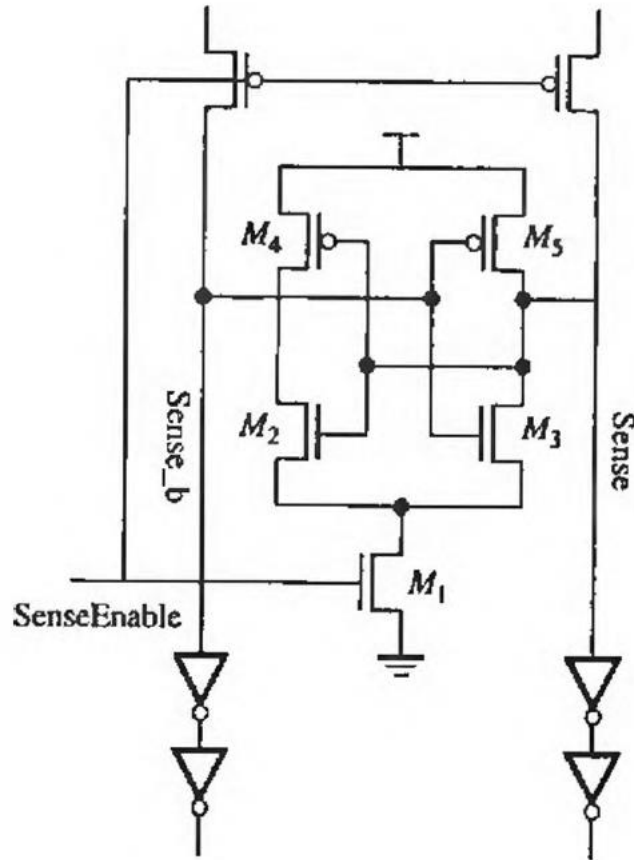
(a) Discharging output



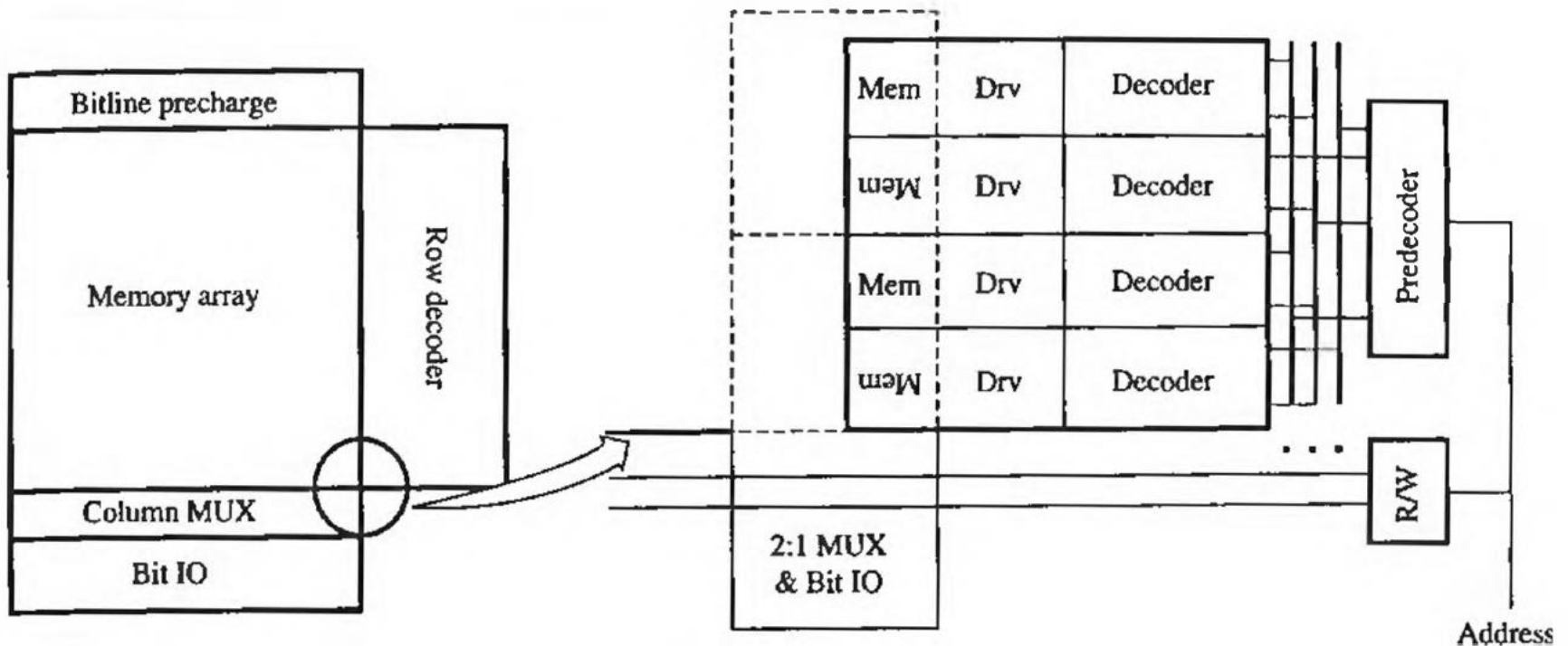
(b) Charging output

- $\Delta V \sim 100$ to 200mV , Speed vs. Power Dissipation
- Slew rate (dV/dt at output) $\frac{dV}{dt} = \frac{I_{SS}}{C_{out}}$
- Delay $\Delta\tau = \frac{C_{out} \Delta V_{out}}{I_{SS}}$
- Power Dissipation $P = I_{SS} V_{DD}$

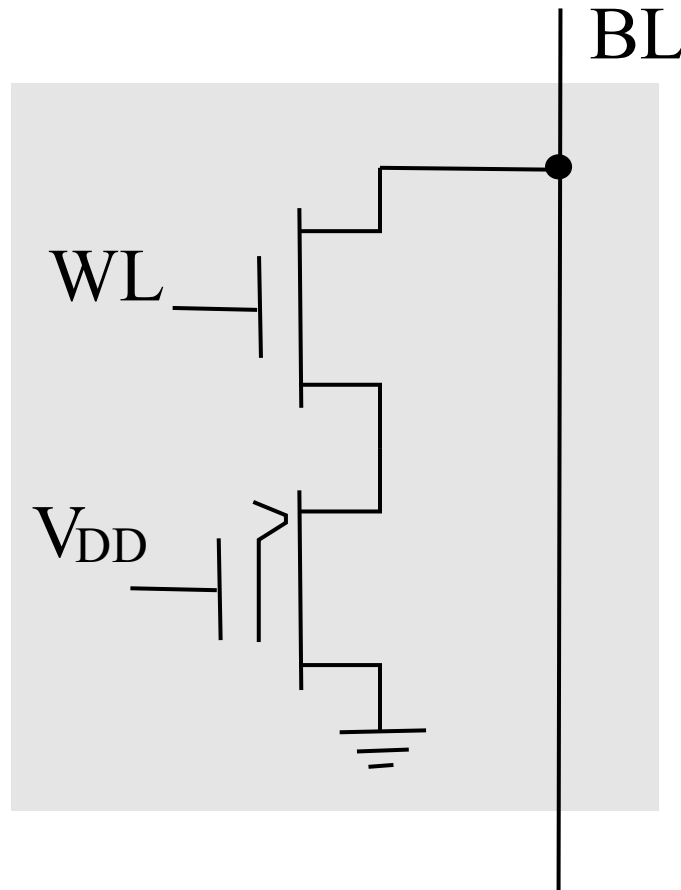
Latch-Based Sense Amplifier



Memory Architecture



EEPROM Cell

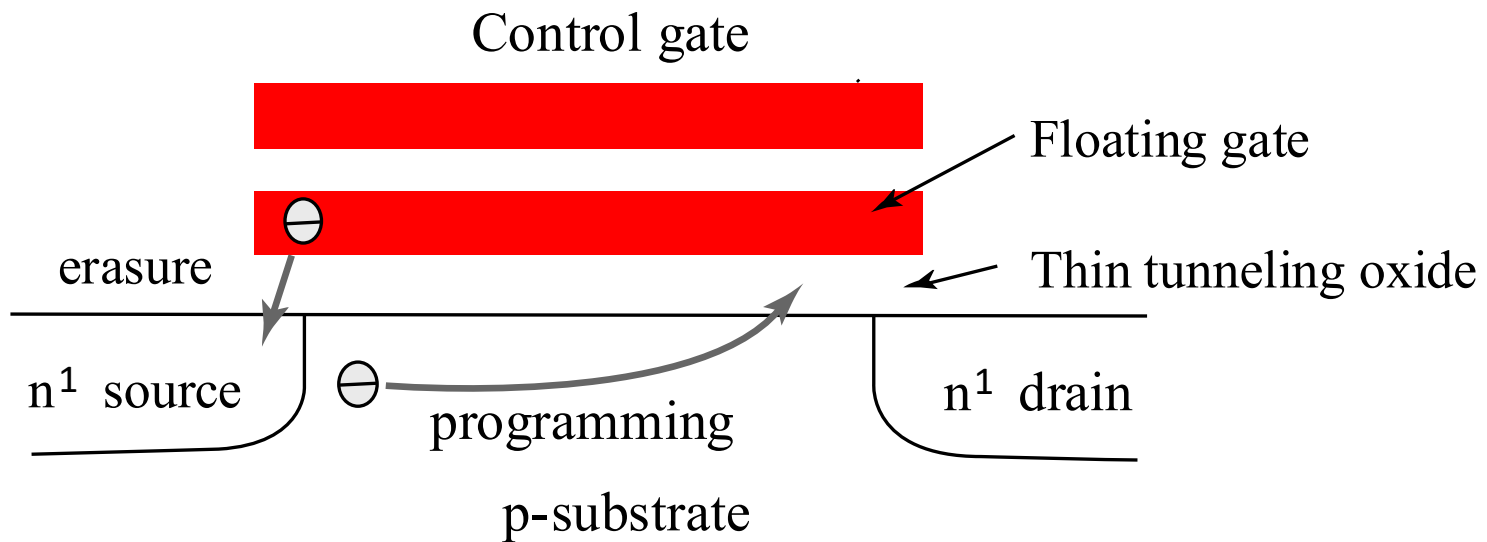


Absolute threshold control
is hard

Unprogrammed transistor
might be depletion

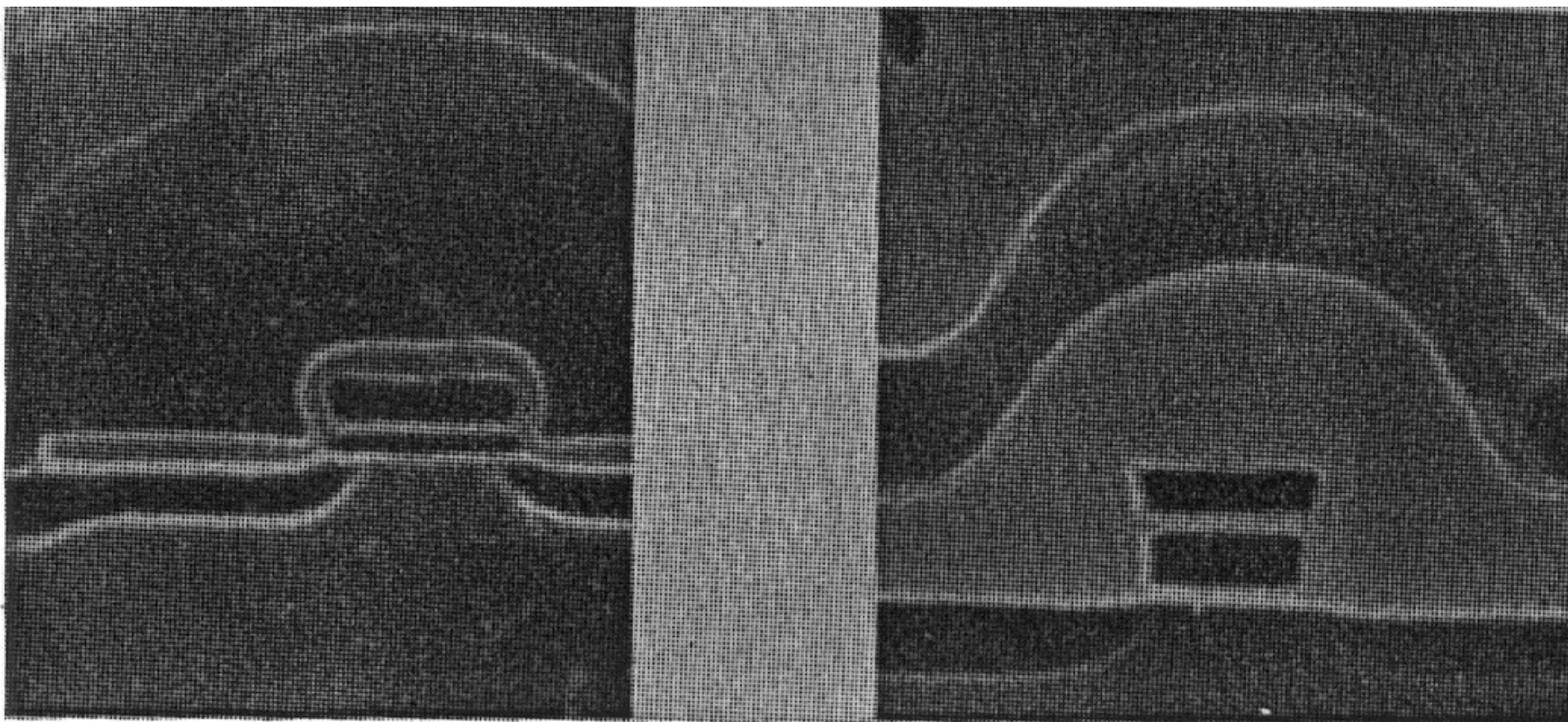
⇒ 2 transistor cell

Flash EEPROM



Many other options ...

Cross-sections of NVM cells



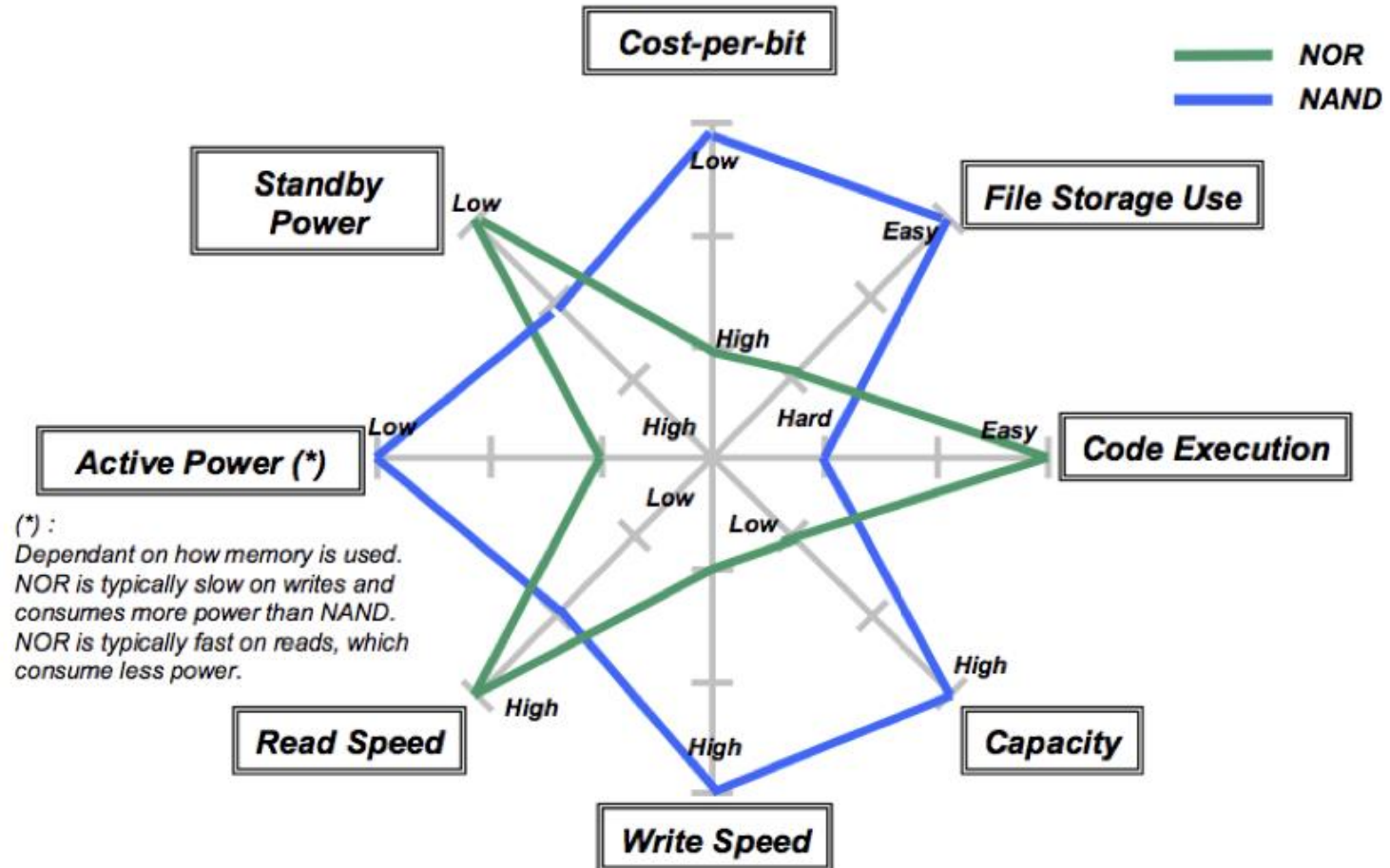
Flash

EPROM

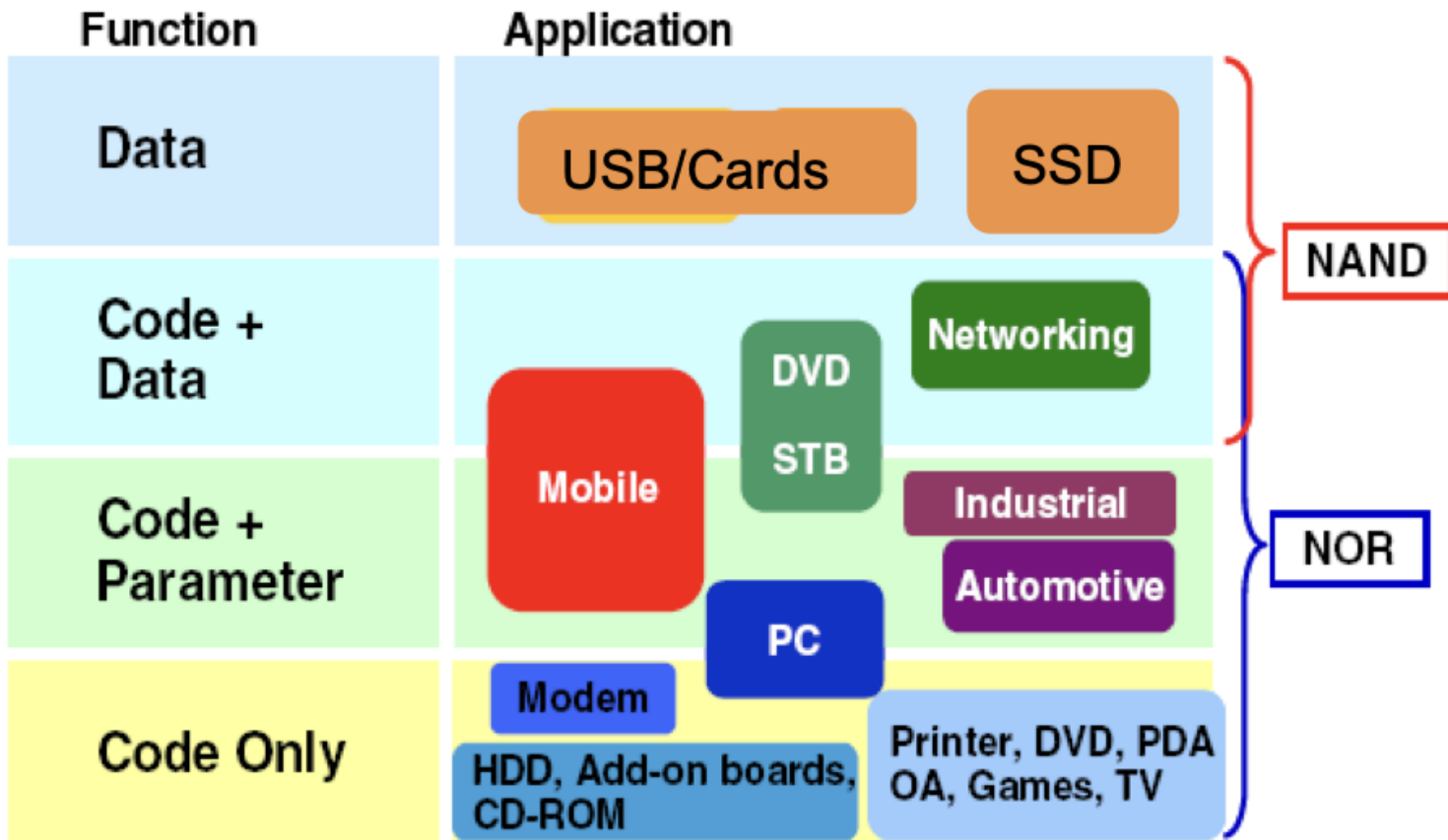
Flash Memories

	NAND	NOR
Cell Array		
Layout		
Cross Section		
Cell Size	$4F^2$	$10F^2$

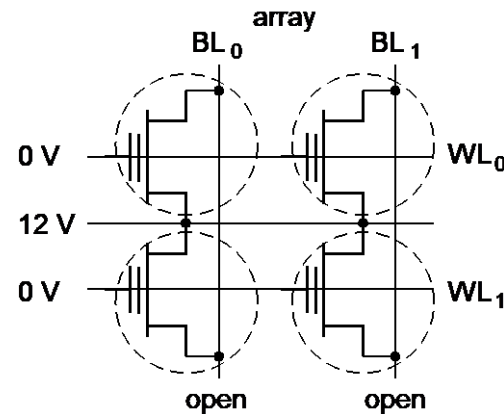
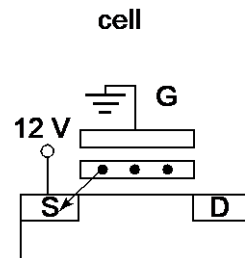
Flash Memories



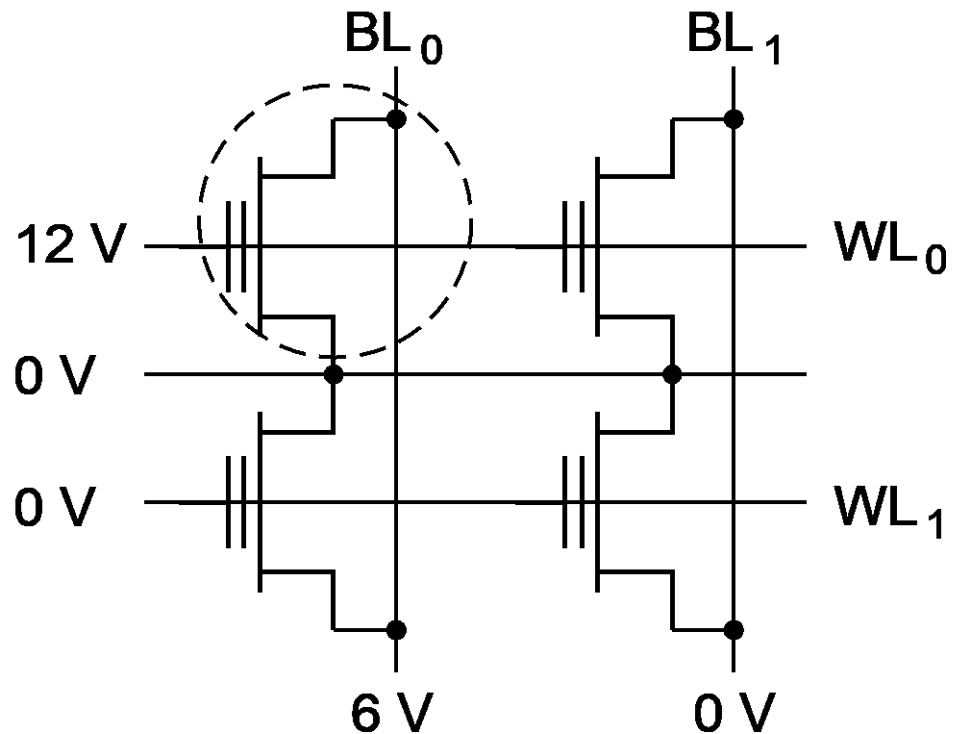
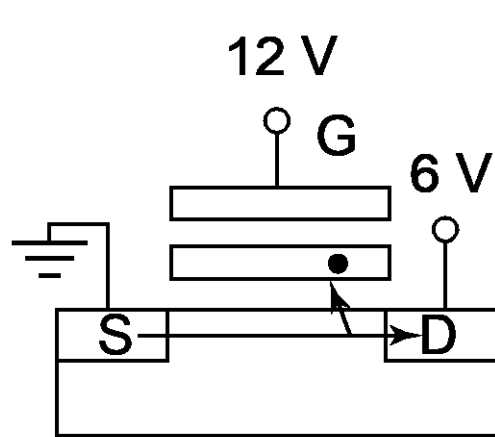
Flash Memories



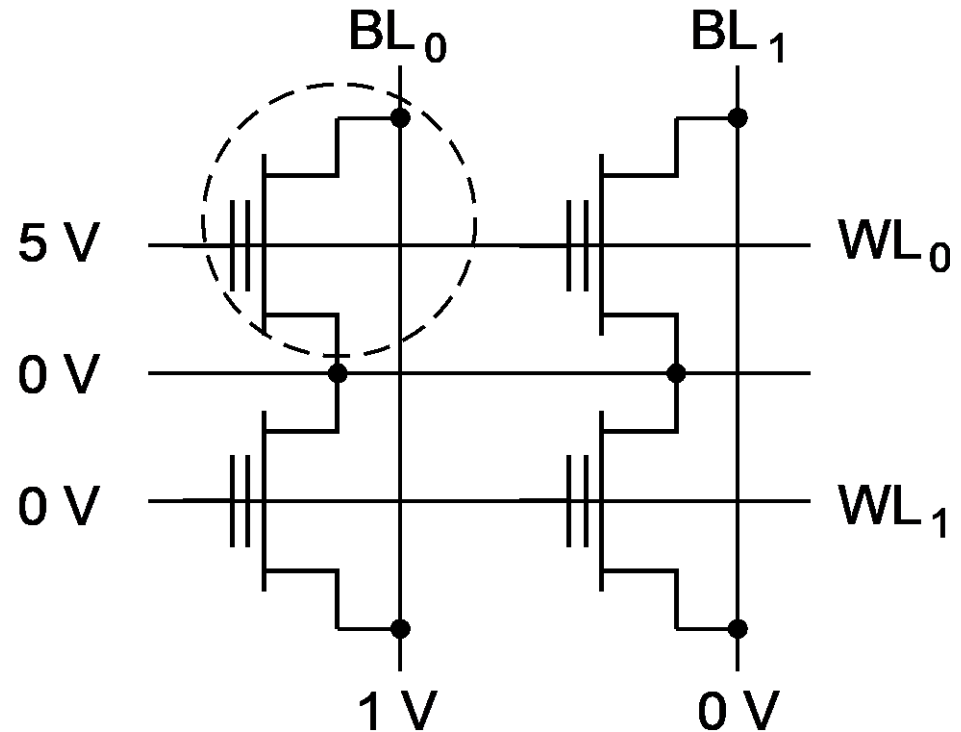
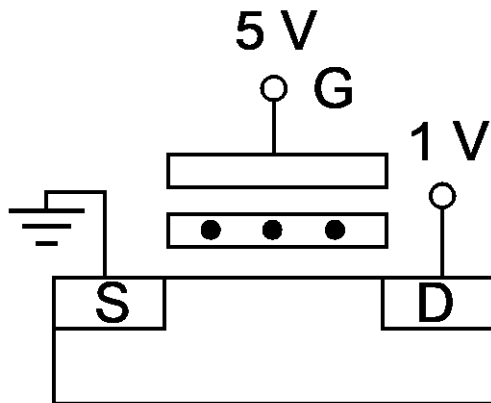
Basic Operations in a NOR Flash Memory— Erase



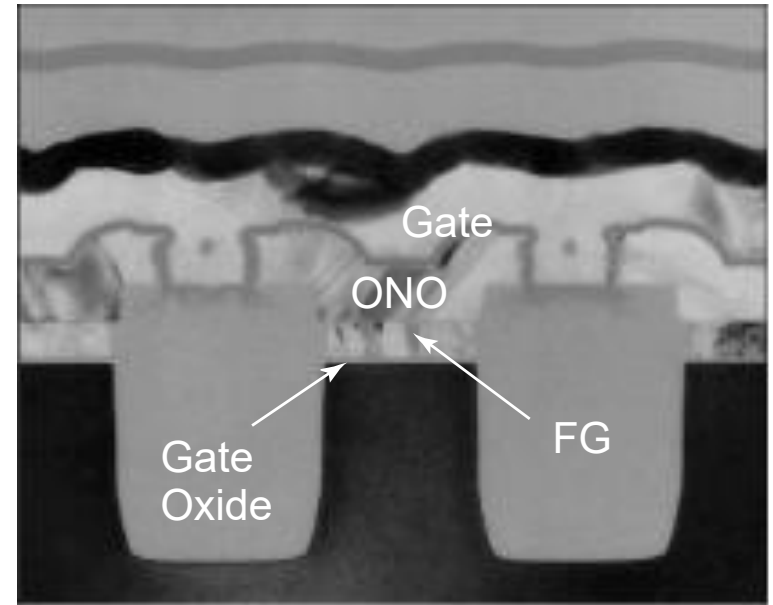
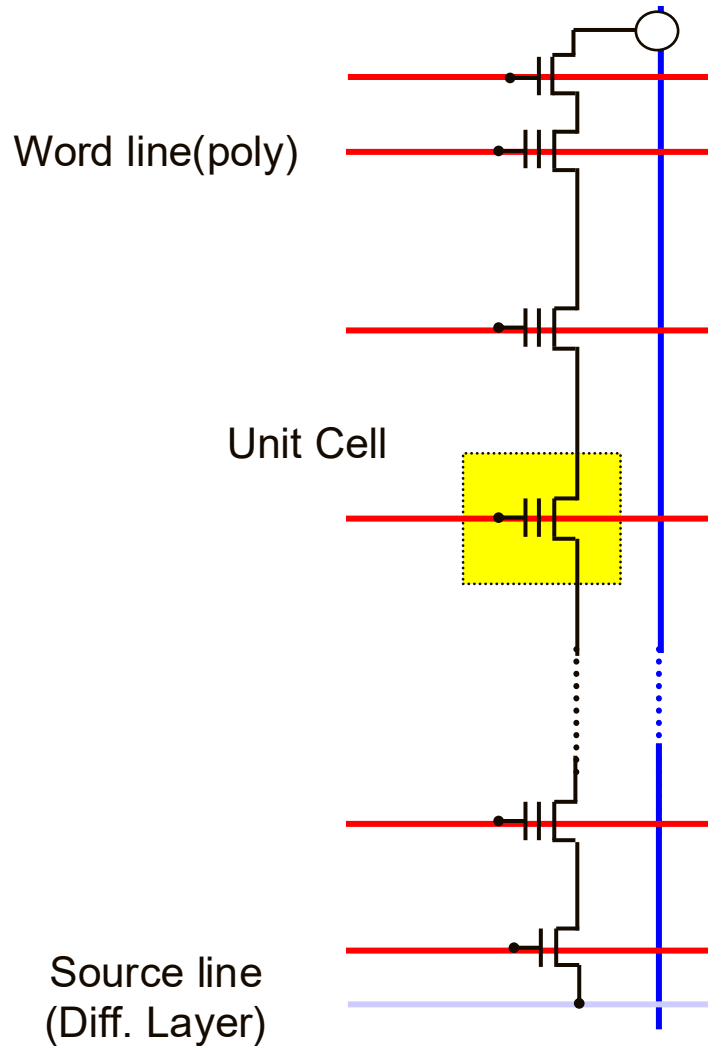
Basic Operations in a NOR Flash Memory— Write



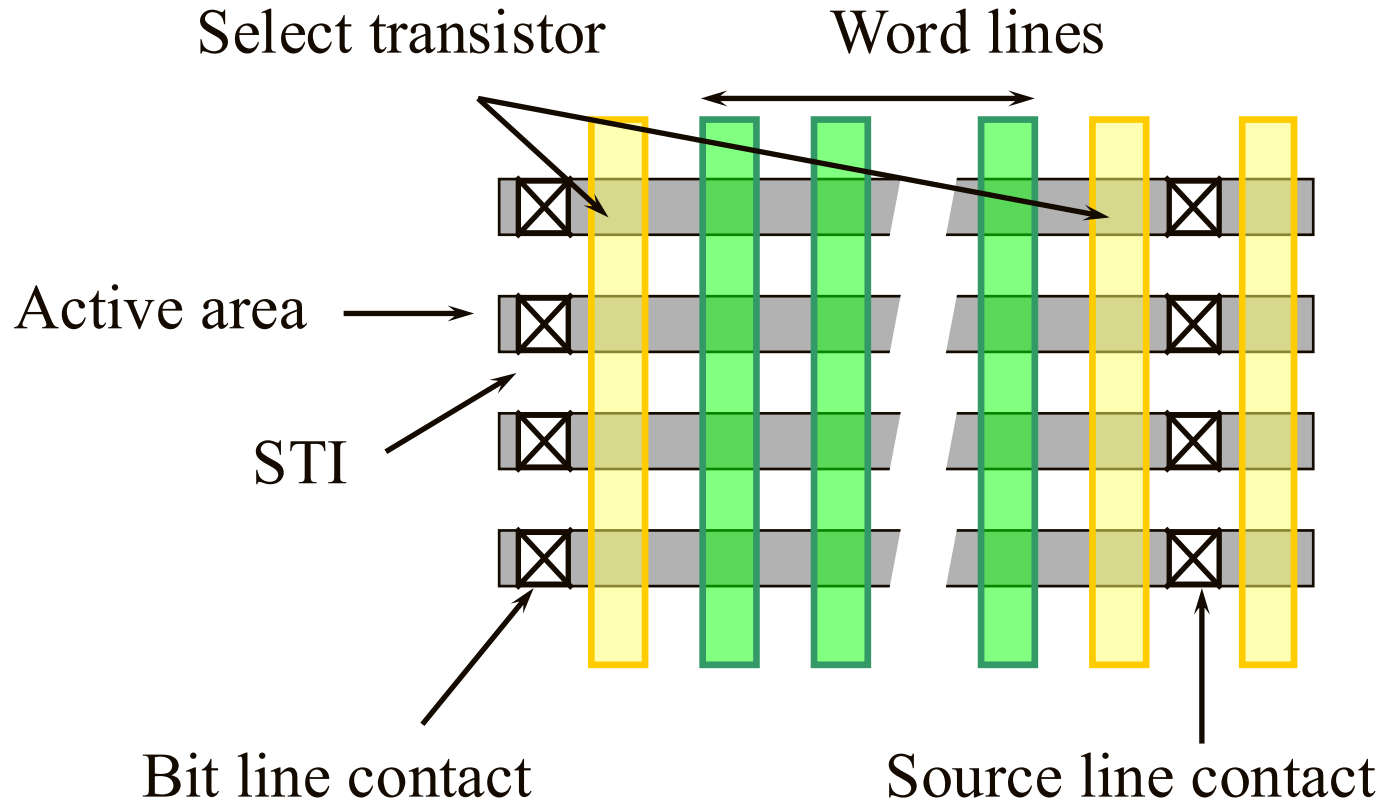
Basic Operations in a NOR Flash Memory— Read



NAND Flash Memory



NAND Flash Memory



Characteristics of State-of-the-art NVM

Table 12-1 Comparison between nonvolatile memories ([Itoh01]).
 $V_{DD} = 3.3$ or 5 V; $V_{PP} = 12$ or 12.5 V.

	Cell— Nr. of Transistors	Cell Area (ratio wrt EPROM)	Mechanism		External Power Supply		Program/ Erase Cycles
			Erase	Write	Write	Read	
MASK ROM	1 T (NAND)	0.35–5	—	—	—	V_{DD}	0
EPROM	1 T	1	UV Exposure	Hot electrons	V_{PP}	V_{DD}	~100
EEPROM	2 T	3–5	FN Tunneling	FN Tunneling	V_{PP} (int)	V_{DD}	10^4 – 10^5
Flash Memory	1 T	1–2	FN Tunneling	Hot electrons	V_{PP}	V_{DD}	10^4 – 10^5
			FN Tunneling	FN Tunneling	V_{PP} (int)	V_{DD}	10^4 – 10^5